

SREE GNANESH DOMMETI

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Education

University of Maryland, College Park

M.S. in Applied Machine Learning; GPA: 3.9 / 4.0

May 2026

College Park, MD

National Institute of Technology, Calicut

B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10

May 2022

Calicut, India

Experience

FROOT Lab, University of Maryland

November 2025 – Present

Research Assistant, Advisor: Alan Zaoring Liu

College Park, MD

- Contributing to ProjectASAP, an approximation-first telemetry research effort that serves approximate analytics from streaming sketches instead of repeated raw time-series scans.
- Built PromQL workload-analysis/sketch-planning paths and NetArena MALT benchmark tooling with Dockerized LiteLLM prompts, output validation, retries, and local submissions.

GAMMA Lab, University of Maryland

January 2026 – May 2026

Research Assistant, Advisor: Dinesh Manocha

College Park, MD

- Contributed to ongoing grounded MLLM/video reasoning evaluation tooling that scores intermediate reasoning steps against visual/textual evidence.

Qualcomm

July 2022 – July 2024

Software Engineer

Hyderabad, India

- Owned LTE Layer 3 RRC Cell Selection and Security modules for 4G/5G, CAT-M1, and NB-IoT modem stacks.
- Built Python log-analysis, regression-tracking, changelist-routing, and fuzzing automation over modem traces, reducing manual triage and assignment effort by 40%.

Connycr

June 2025 – August 2025

ML Engineer Intern, CampusAI Event Recommendation

College Park, MD

- Built a personalized campus event recommendation system with FastAPI, Elasticsearch lexical retrieval, MiniLM semantic search, and reciprocal-rank fusion; achieved sub-300 ms p95 retrieval and sub-600 ms p95 end-to-end latency.
- Implemented ALS/BPR collaborative filtering and offline ranking evaluation over implicit interactions, tracking Recall@K, NDCG@K, MRR@K, cold-start recall, long-tail coverage, and RSVP/save-rate proxy metrics with MLflow.

Projects

DistTrain Lab - LLM Pretraining & Post-Training Lab | *Python, TRL/Accelerate, FSDP, DeepSpeed*

- Added causal-LM pretraining plans for an 8x L40 target with data-curation manifests, LR/batch/precision sweeps, validation-loss/perplexity gates, checkpoint checks, deterministic smoke runs, and DeepSpeed/FSDP/Megatron/TorchTitan-style configs.

Benchmarking LLM Inference Frameworks | *Python, PyTorch, vLLM, SGLang, Docker, AWS EC2 GPU*

- Deployed vLLM and SGLang to serve Mistral-7B-Instruct-AWQ on EC2 GPU; benchmarked 1–64 concurrent users for latency, throughput, cost, GPU utilization, and saturation behavior.

CallIntel Agent - Audio Support-Call QA | *Python, GCP, AWS Bedrock, OpenTelemetry, Datadog*

- Built an audio-to-RAG support QA prototype with PII redaction, cited answers, Gemini/OpenAI/Claude/Bedrock payload validation, GCP artifacts, and Datadog-compatible OpenTelemetry quality/latency metrics.

Real-Time Fraud Detection ML Lifecycle | *PyTorch, Airflow, dbt/Snowflake, Databricks, AWS ECS*

- Extended a fraud classifier into an ML lifecycle pipeline with DVC/ClearML tracking, drift reports, promotion gates, Dockerized FastAPI inference, Airflow orchestration, dbt/Snowflake marts, and Databricks job specs.

Skills

Languages: Python, SQL, C, C++, Rust

ML / GenAI: PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace, XGBoost, LightGBM, LangChain, LangGraph, LlamaIndex, RAG, causal-LM pretraining workflows, Gemini JSON APIs, AWS Bedrock payloads, LoRA, DPO, PPO/GRPO-style rewards, vLLM, SGLang

Distributed Training: TRL, Accelerate, Ray, DeepSpeed, FSDP, Megatron/TorchTitan configs, data/tensor/pipeline parallelism, activation checkpointing, 8-GPU L40 launch configs

Data / Infra: Kafka, Arroyo, Spark, Databricks/Spark job specs, Snowflake/dbt artifacts, Redis, PostgreSQL, Elasticsearch, FAISS, Pinecone, Prometheus, Grafana, OpenTelemetry/Datadog-compatible OTLP

MLOps: Docker, Kubernetes, Airflow DAGs, AWS ECS/EC2/S3/SageMaker/Lambda, GCP Cloud Run/Pub/Sub/BigQuery/IAM, Terraform, MLflow, ClearML, DVC, W&B, GitHub Actions, EvidentlyAI